

Generation Interconnection System Impact Study Report

For Queue Project Q21-1817

Waterford Power Llc 230 kV

January 20, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1817. The Interconnection Customer proposes a 639.4 MW natural gas combined cycle facility interconnecting at the Waterford Power Llc 230 kV substation in the Ohio area. The request was received on July 18, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 534 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,674,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Waterford Power Llc / Muskingum Riv 230 KV	Myrtle-Date / Waterford Power Llc 230 KV line	102.0	460	62.4
Brentwood / Muskingum Riv 230 KV Line	Tidd / Muskingum Riv 500 KV line	100.7	580	10.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Waterford Power Llc / Muskingum Riv 230 KV	Reconductor line	\$2,311,000	100% (\$2,311,000)
Brentwood / Muskingum Riv 230 KV Line	Reconductor line	\$19,363,000	100% (\$19,363,000)
	Total allocated to project		\$21,674,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.